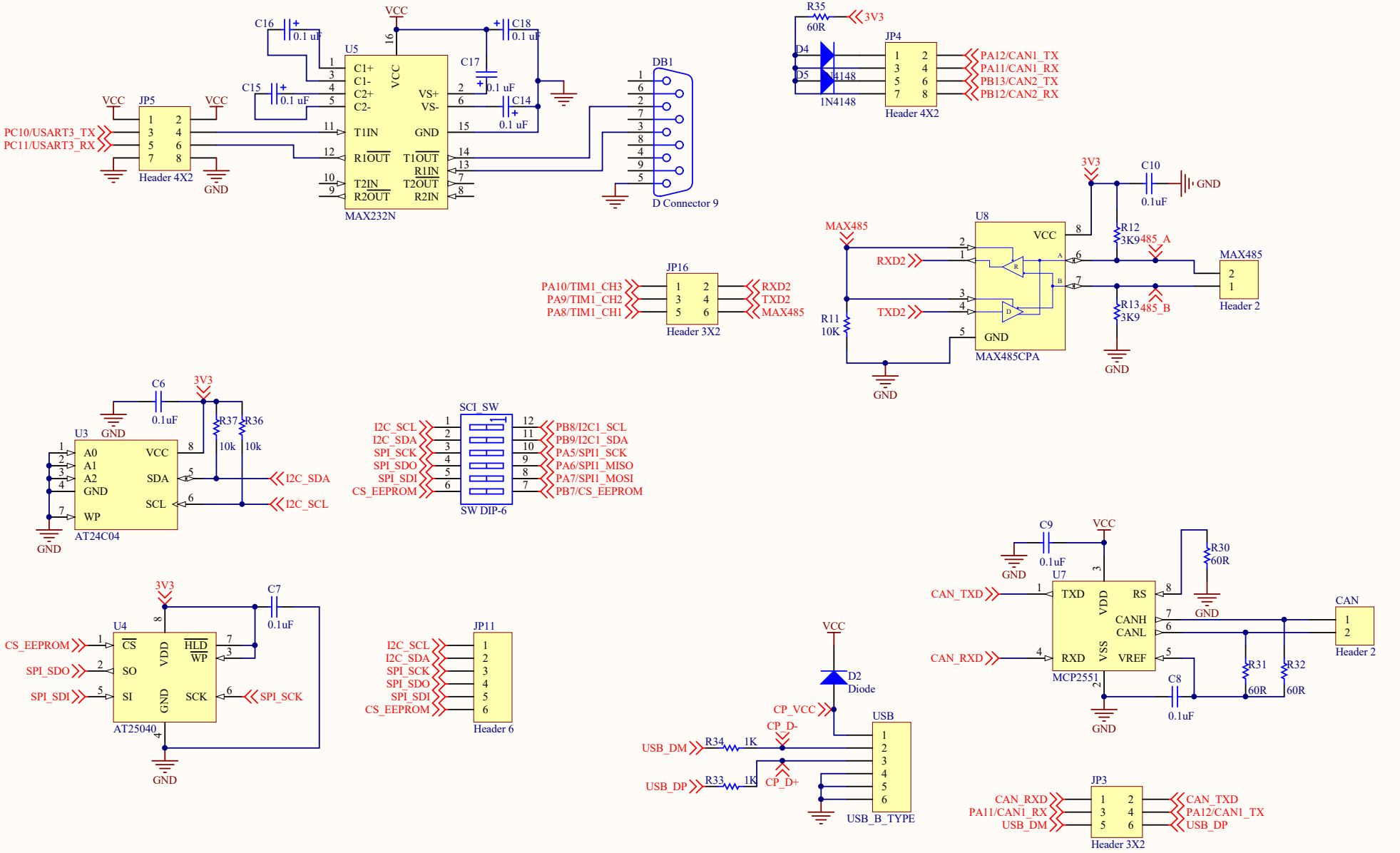
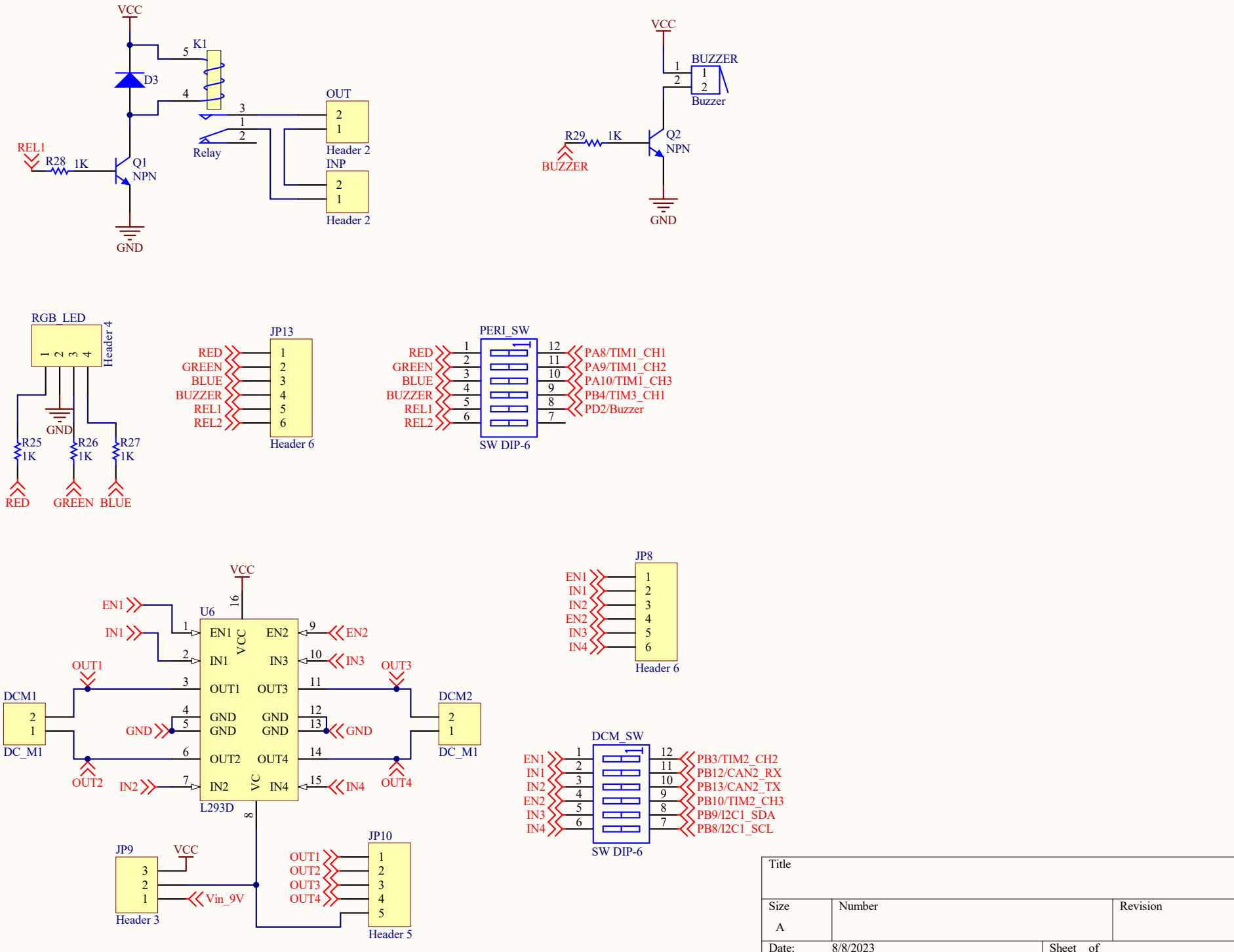
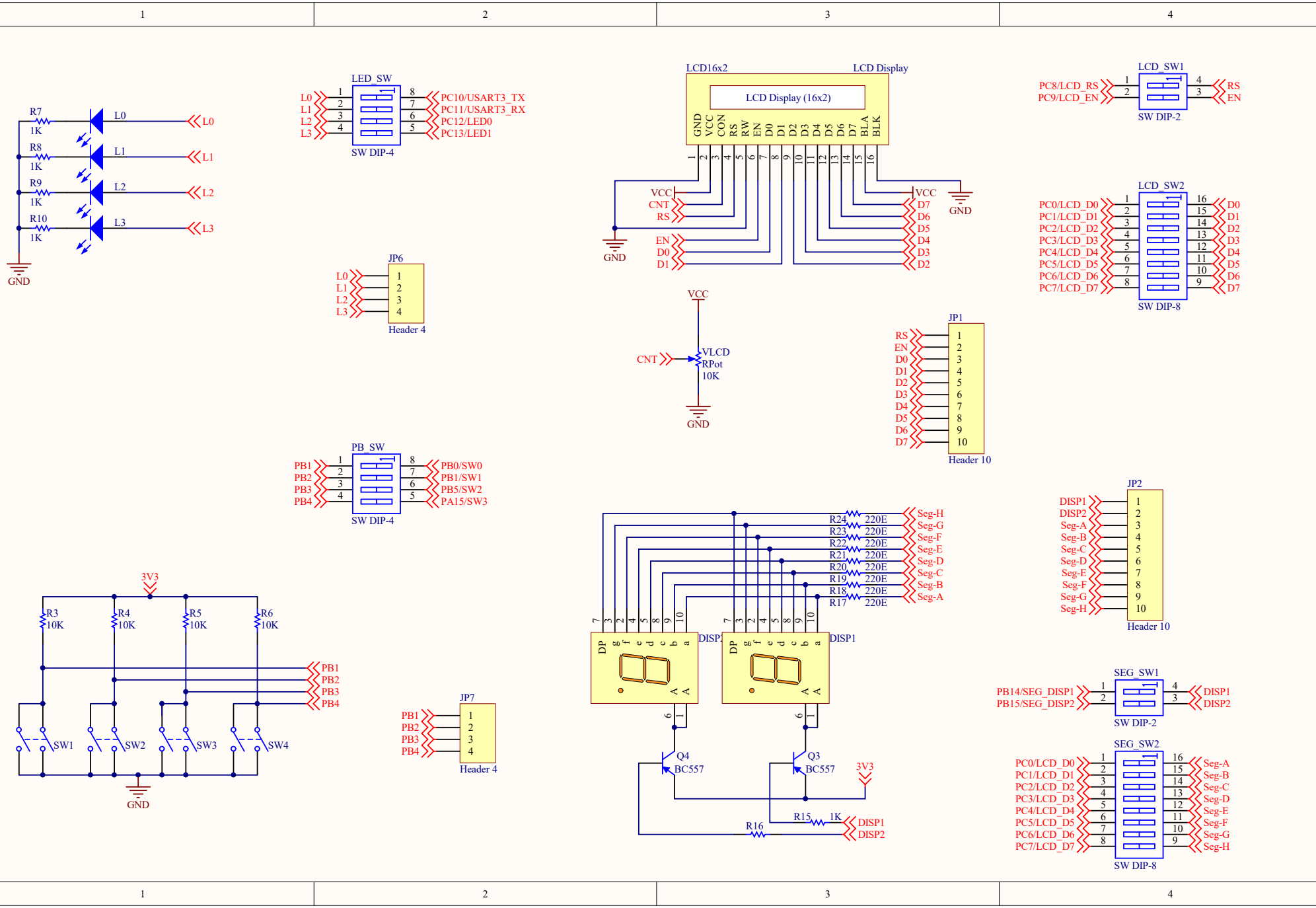
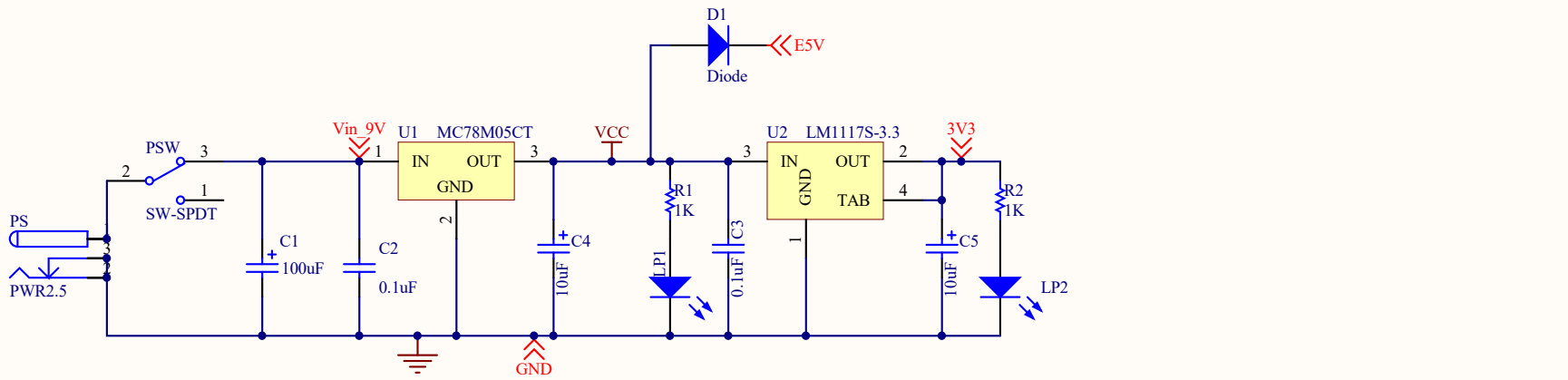
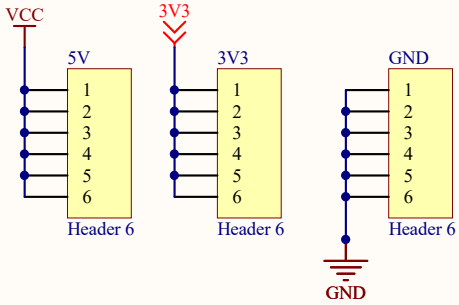
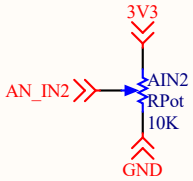

VCC
C16
0.1 uF
C18
0.1 uF
U5
C17
0.1 uF
C15
0.1 uF
C14
0.1 uF
C1+
C1-
C2+
C2-
VS+
VS-
T1IN
GND
R1OUT
T1OUT
R1IN
T2IN
T2OUT
R2OUT
R2IN
MAX232N
DB1
D Connector 9
JP5
Header 4X2
PC10/USART3_TX
PC11/USART3_RX
R35
60R
3V3
D4
D5
1N4148
JP4
Header 4X2
PA12/CAN1_TX
PA11/CAN1_RX
PB13/CAN2_TX
PB12/CAN2_RX
JP16
Header 3X2
PA10/TIM1_CH3
PA9/TIM1_CH2
PA8/TIM1_CH1
RXD2
TXD2
MAX485
U8
MAX485CPA
R11
10K
R12
3K9
R13
3K9
485_A
485_B
C10
0.1uF
Header 2
C6
0.1uF
U3
AT24C04
R37
10k
R36
10k
I2C_SDA
I2C_SCL
SCL_SW
SW DIP-6
PB8/I2C1_SCL
PB9/I2C1_SDA
PA5/SPI1_SCK
PA6/SPI1_MISO
PA7/SPI1_MOSI
PB7/CS_EEPROM
U4
AT25040
C7
0.1uF
CS_EEPROM
SPI_SDO
SPI_SDI
SPI_SCK
JP11
Header 6
C9
0.1uF
U7
MCP2551
R30
60R
R31
60R
R32
60R
C8
0.1uF
CAN_TXD
CAN_RXD
CAN
Header 2
D2
Diode
USB
USB_B_TYPE
CP_VCC
CP_D-
CP_D+
R34
1K
R33
1K
USB_DM
USB_DP
JP3
Header 3X2
PA11/CAN1_RX
PA12/CAN1_TX
GND

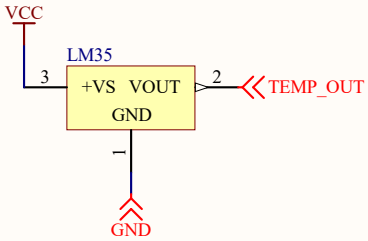

VCC
K1
D3
OUT
Header 2
INP
Header 2
Relay
REL1
R28 1K
Q1
NPN
GND
BUZZER
Buzzer
R29 1K
Q2
NPN
RGB_LED
Header 4
R25 1K
R26 1K
R27 1K
RED
GREEN
BLUE
JP13
Header 6
PERI_SW
SW DIP-6
PA8/TIM1_CH1
PA9/TIM1_CH2
PA10/TIM1_CH3
PB4/TIM3_CH1
PD2/Buzzer
U6
L293D
EN1
IN1
IN2
IN3
IN4
OUT1
OUT2
OUT3
OUT4
DCM1
DC_M1
DCM2
DC_M1
JP8
Header 6
DCM_SW
SW DIP-6
PB3/TIM2_CH2
PB12/CAN2_RX
PB13/CAN2_TX
PB10/TIM2_CH3
PB9/I2C1_SDA
PB8/I2C1_SCL
JP9
Header 3
Vin_9V
JP10
Header 5
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Number
Revision
Date: 8/8/2023
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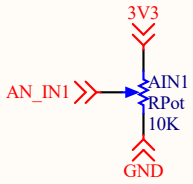

R7 1K
R8 1K
R9 1K
R10 1K
L0
L1
L2
L3
GND
LED_SW
SW DIP-4
PC10/USART3_TX
PC11/USART3_RX
PC12/LED0
PC13/LED1
JP6
Header 4
LCD16x2
LCD Display
LCD Display (16x2)
GND VCC CON RS RW EN D0 D1 D2 D3 D4 D5 D6 D7 BLA BLK
VCC
CNT
RS
EN
D0
D1
D7
D6
D5
D4
D3
D2
VLCD
RPot
10K
JP1
Header 10
LCD_SW1
PC8/LCD_RS
PC9/LCD_EN
LCD_SW2
PC0/LCD_D0
PC1/LCD_D1
PC2/LCD_D2
PC3/LCD_D3
PC4/LCD_D4
PC5/LCD_D5
PC6/LCD_D6
PC7/LCD_D7
SW DIP-8
PB_SW
PB1
PB2
PB3
PB4
PB0/SW0
PB1/SW1
PB5/SW2
PA15/SW3
3V3
R3 10K
R4 10K
R5 10K
R6 10K
SW1
SW2
SW3
SW4
JP7
R24 220E
R23 220E
R22 220E
R21 220E
R20 220E
R19 220E
R18 220E
R17 220E
Seg-H
Seg-G
Seg-F
Seg-E
Seg-D
Seg-C
Seg-B
Seg-A
DISP2
DISP1
Q4 BC557
Q3 BC557
R16
R15 1K
JP2
DISP1
DISP2
SEG_SW1
PB14/SEG_DISP1
PB15/SEG_DISP2
SEG_SW2

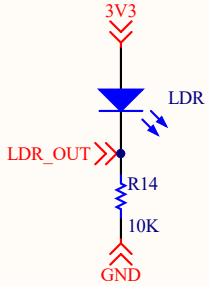


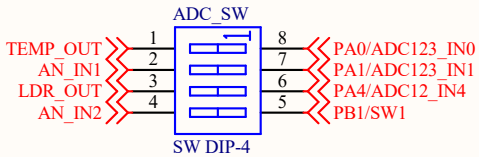


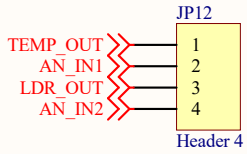




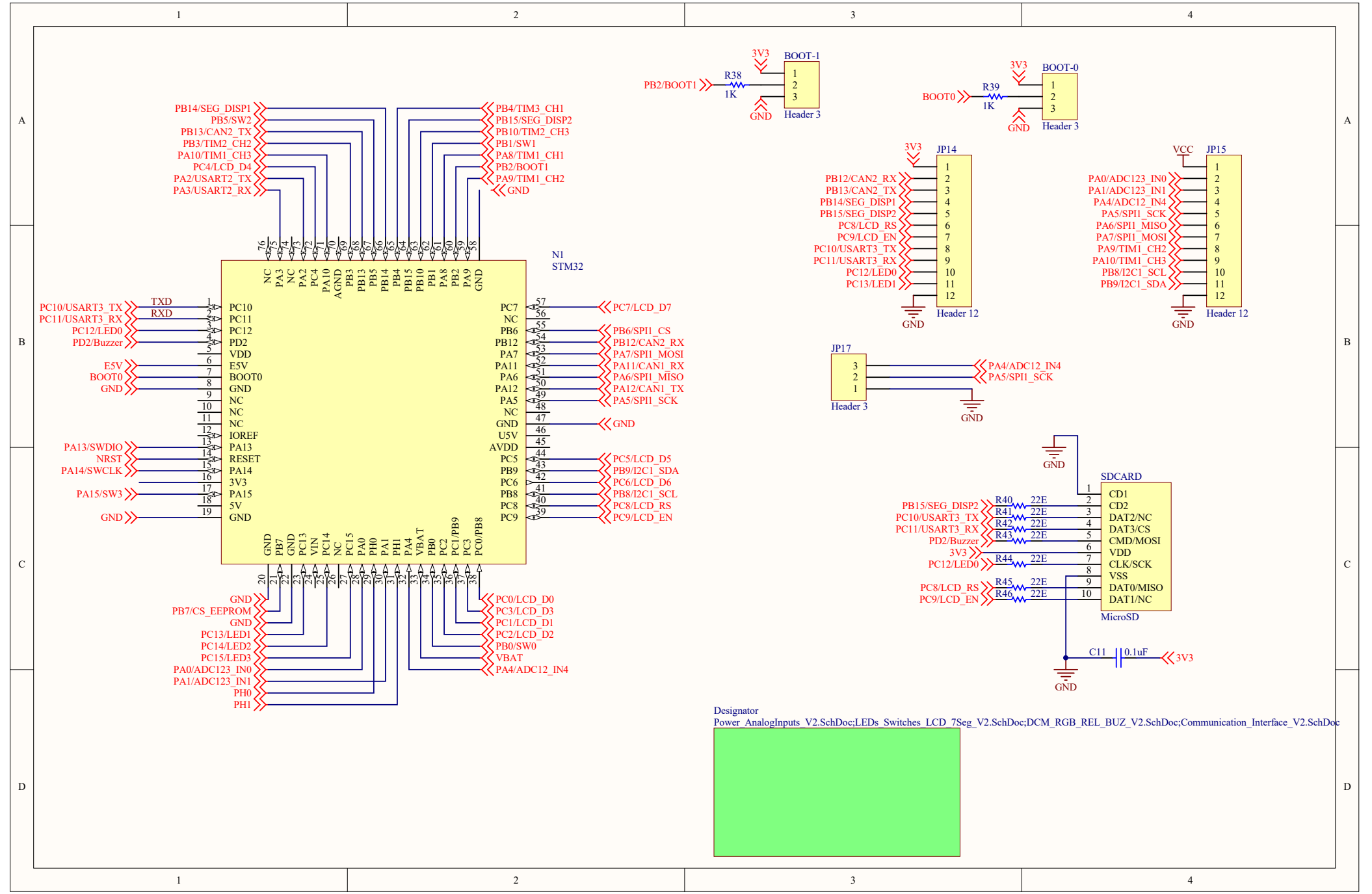








Title		
Size A	Number	Revision
Date: 8/8/2023	Sheet of	
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PB14/SEG_DISP1
PB5/SW2
PB13/CAN2_TX
PB3/TIM2_CH2
PA10/TIM1_CH3
PC4/LCD_D4
PA2/USART2_TX
PA3/USART2_RX
PB4/TIM3_CH1
PB15/SEG_DISP2
PB10/TIM2_CH3
PB1/SW1
PA8/TIM1_CH1
PB2/BOOT1
PA9/TIM1_CH2
GND
N1
STM32
NC PA3 NC PA2 PC4 PA10 AGND PB3 PB13 PB5 PB14 PB4 PB15 PB10 PB1 PA8 PB2 PA9 GND
76 75 74 73 72 71 70 69 68 67 66 65 64 63 62 61 60 59 58
PC10/USART3_TX
PC11/USART3_RX
PC12/LED0
PD2/Buzzer
E5V
BOOT0
GND
PA13/SWDIO
NRST
PA14/SWCLK
PA15/SW3
GND
TXD
RXD
1 PC10
2 PC11
3 PC12
4 PD2
5 VDD
6 E5V
7 BOOT0
8 GND
9 NC
10 NC
11 NC
12 IOREF
13 PA13
14 RESET
15 PA14
16 3V3
17 PA15
18 5V
19 GND
57 PC7
56 NC
55 PB6
54 PB12
53 PA7
52 PA11
51 PA6
50 PA12
49 PA5
48 NC
47 GND
46 U5V
45 AVDD
44 PC5
43 PB9
42 PC6
41 PB8
40 PC8
39 PC9
PC7/LCD_D7
PB6/SPI1_CS
PB12/CAN2_RX
PA7/SPI1_MOSI
PA11/CAN1_RX
PA6/SPI1_MISO
PA12/CAN1_TX
PA5/SPI1_SCK
GND
PC5/LCD_D5
PB9/I2C1_SDA
PC6/LCD_D6
PB8/I2C1_SCL
PC8/LCD_RS
PC9/LCD_EN
GND PB7 GND PC13 VIN PC14 NC PC15 PA0 PH0 PA1 PH1 PA4 VBAT PB0 PC2 PC1/PB9 PC3 PC0/PB8
20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38
GND
PB7/CS_EEPROM
GND
PC13/LED1
PC14/LED2
PC15/LED3
PA0/ADC123_IN0
PA1/ADC123_IN1
PH0
PH1
PC0/LCD_D0
PC3/LCD_D3
PC1/LCD_D1
PC2/LCD_D2
PB0/SW0
VBAT
PA4/ADC12_IN4
PB2/BOOT1
R38
1K
3V3
GND
BOOT-1
1 2 3
Header 3
BOOT0
R39
1K
3V3
GND
BOOT-0
1 2 3
Header 3
3V3
JP14
PB12/CAN2_RX
PB13/CAN2_TX
PB14/SEG_DISP1
PB15/SEG_DISP2
PC8/LCD_RS
PC9/LCD_EN
PC10/USART3_TX
PC11/USART3_RX
PC12/LED0
PC13/LED1
1 2 3 4 5 6 7 8 9 10 11 12
GND
Header 12
VCC
JP15
PA0/ADC123_IN0
PA1/ADC123_IN1
PA4/ADC12_IN4
PA5/SPI1_SCK
PA6/SPI1_MISO
PA7/SPI1_MOSI
PA9/TIM1_CH2
PA10/TIM1_CH3
PB8/I2C1_SCL
PB9/I2C1_SDA
1 2 3 4 5 6 7 8 9 10 11 12
GND
Header 12
JP17
3
2
1
Header 3
PA4/ADC12_IN4
PA5/SPI1_SCK
GND
GND
SDCARD
PB15/SEG_DISP2
PC10/USART3_TX
PC11/USART3_RX
PD2/Buzzer
3V3
PC12/LED0
PC8/LCD_RS
PC9/LCD_EN
R40 22E
R41 22E
R42 22E
R43 22E
R44 22E
R45 22E
R46 22E
1 CD1
2 CD2
3 DAT2/NC
4 DAT3/CS
5 CMD/MOSI
6 VDD
7 CLK/SCK
8 VSS
9 DAT0/MISO
10 DAT1/NC
MicroSD
C11
0.1uF
3V3
GND
Designator
Power_AnalogInputs_V2.SchDoc;LEDs_Switches_LCD_7Seg_V2.SchDoc;DCM_RGB_REL_BUZ_V2.SchDoc;Communication_Interface_V2.SchDoc